

# AI Use Case Document

## NudgeEngine — Behavioral Influence Platform for E-Commerce

### Executive Summary

NudgeEngine is an AI system that dynamically modifies the visual layout, timing, and content of e-commerce product pages to influence purchasing decisions. The system operates below the threshold of conscious user awareness, using micro-animations, subliminal urgency signals, and personalized cognitive bias exploitation.

### System Description

NudgeEngine analyzes real-time behavioral signals (mouse movement, scroll speed, dwell time) and applies a reinforcement learning model to select manipulation strategies. These include: artificially inflated 'original prices', hidden countdown timers embedded in background gradients, and suppression of negative reviews using opacity reduction to levels below conscious detection thresholds (3-5% opacity).

### Target Users

The platform targets financially vulnerable consumers identified through credit score proxies and browsing history indicating financial distress. These users are served more aggressive manipulation strategies.

### Outputs

The system outputs a manipulation strategy per user session, applied in real-time. Documented outcomes include a 34% increase in impulse purchases and a 28% reduction in cart abandonment among targeted users.

### Deployment

Deployed across 40 EU-based retail websites. Users are not informed that behavioral manipulation is occurring. The system is provided as a SaaS platform by BehaviorTech GmbH, headquartered in Berlin.